Data Science

Model Explainability

GENERAL ASSEMBLY

Lesson roadmap

- Welcome + Introduction
- Explainability
- The Importance of Explainability
- White Box vs Black Box Models
- Building Explainable Models
- Bring It Home

Learning objectives

1. Understand what model interpretability is, and why it's important.
2. Discuss real life case studies of more and less explainable models
3. Understand and implement metrics including feature importance and decision tree visualisations as examples of explainability.

What is "Explainability"?

Explainability

What is explainability?

- Imagine an AI model was used to assess your credit score
- This affects your ability to rent an apartment, take out loans, and get a mortgage
- Your score comes out much lower than you expected
- What questions would you have about this process?

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What is model explainability?

In the previous scenario, you would probably want to know:

- How did the model calculate your credit score?
- Which factors did it take into account?
- What weighting did it give to these different factors?

All of these questions are examples of what makes a model explainable or interpretable.

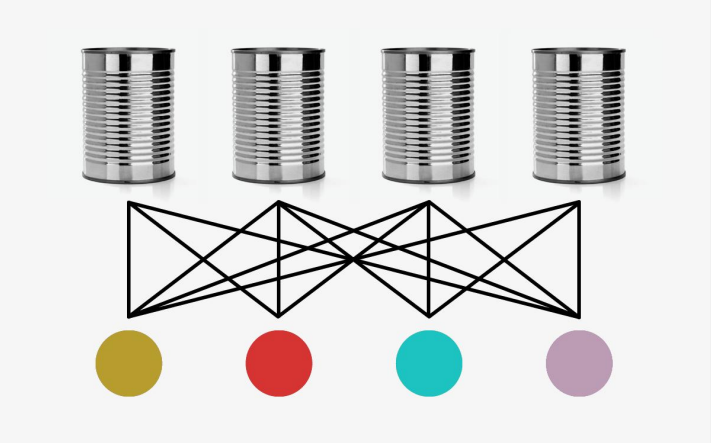
Source

What is model explainability?

Model explainability is the ability to understand how a model makes its decisions.

It allows us to answer questions like...

- Why did the model make this decision?
- What factors were most important in the model's decision-making process?
- Can I trust the model's output?
- What would happen if the input data changed slightly?



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What is model explainability?

Let's look at three important components of explainability.

TRANSPARENCY	INTERPRETABILITY	FEATURE IMPORTANCE
How easy is it to understand the model's decision making process? What are the relationships between model inputs and outputs?	Can the model's decision making process be explained in human-friendly terms? For example "The customer was denied a loan because their income was below the threshold"	Which features were most important when making the decision? What is the relative weighting given to different features?

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The Importance of Explainability

Why is explainability important?

TRUST	FAIRNESS	REGULATION	MODEL IMPROVEMENT	ETHICAL RISK REDUCTION
Users are more likely to trust and use predictive models if they are transparent and explainable.	More explainable models are easier to audit for bias.	In some parts of the world, in fields such as healthcare or finance, decisions made by predictive models must be explainable. For example, the "right to an explanation" is included in GDPR.	Understanding how a model makes its decisions helps data scientists to identify and troubleshoot issues, and then refine their models progressively for better performance.	For AI models, explainability allows proactive identification of potential copyright risks before deployment, minimizing reputational and legal damage.

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Model Explainability

Do all models need to be explainable?

- Some models are inherently more explainable than others (we'll discuss this in detail later).

- Is this necessarily a bad thing?
- Are there some scenarios where explainability is less important?

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Do all models need to be explainable?

Explainability is most critical when models are used:	Explainability is less critical when models are used:
• To make decisions that directly impact human health or livelihood, for example healthcare and finance. • In fields where an explanation is required by law- for example GDPR. • A high degree of trust and ethical standards are needed- for example policy making.	• In applications where the outcome is more important than the decision making process, for example image recognition. • In fast paced applications where providing real time explanations is impractical.

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Explainability in credit scoring

- ☐ Take some time to read this article from the LA Times.
- ☐ It describes a startup which uses many different, sometimes unexpected, variables to predict creditworthiness.

- ☐ Why is explainability important in this instance?
- ☐ Is this model explainable?
- ☐ The CEO says that handwriting is correlated with creditworthiness. Is it a good idea to use this as a variable in the model?

Source

Explainability in credit scoring

- This model is not explainable.
- The CEO cannot explain why some features are strongly correlated to creditworthiness, including handwriting: “we don’t know why, it just is”.
- The model uses many features, and it is unclear what weighting the model gives to them.
- It is unclear how the model makes its decisions.

Los Angeles Times

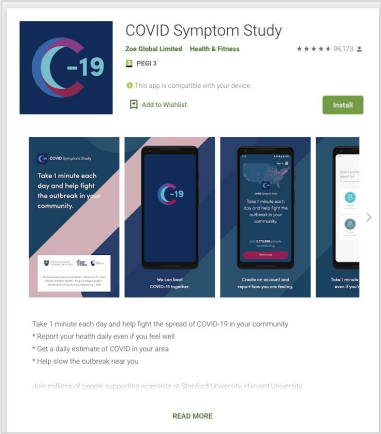
BUSINESS

Some lenders are judging you on much more than finances

Source

Explainability in healthcare

- ☐ Researchers at King's College London built a model that predicts the chances that a person has covid, based on their symptoms.
- ☐ Read through the Nature paper that describes this model.



- ☐ Can you find the equation that describes the model?
- ☐ Is this an explainable model?
- ☐ What can we say about how the model makes decisions?

Source

Explainability in healthcare

- This model is simple and highly explainable, but still powerful and useful.
- The features, and their relative weights, are clear.
- This model would be easy for the researchers, domain experts (clinicians) and end users to interpret and sense check.

$$\text{Prediction model} = -1.32 - (0.01 \times \boxed{\text{age}}) + (0.44 \boxed{\times \text{sex}}) + (1.75 \times \boxed{\text{loss of smell and taste}}) + (0.31 \times \boxed{\text{severe or significant persistent cough}}) + (0.49 \times \boxed{\text{severe fatigue}}) + (0.39 \times \boxed{\text{skipped meals}})$$

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White Box vs Black Box Models

White box vs black box

Take a look at the models listed below.

Are these more explainable (white box) or less explainable (black box)? Why?

Deep neural networks

Linear regression

Decision trees

Random forests

Logistic regression

Support vector machines

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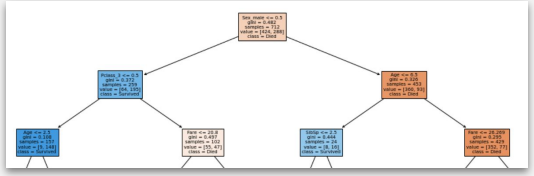
White Box Models

Model	Why is it white box?
Linear regression	Uses a simple mathematical equation, where the relationship between the independent and dependent variables is clear.
Logistic regression	Similar to linear regression, the output log-odds ratio is a weighted sum of independent variables.
Decision trees	The decision making process follows an intuitive tree structure, which is easy to visualise.



$$\text{Prediction model} = -1.32 - (0.01 \times \text{age}) + (0.44 \times \text{sex}) + (1.75 \times \text{loss of smell and taste}) + (0.31 \times \text{severe or significant persistent cough}) + (0.49 \times \text{severe fatigue}) + (0.39 \times \text{skipped meals})$$

Black Box Models

Model	Why is it black box?
Deep neural network *	There are many layers of neurons, each performing complex calculations which are difficult to interpret in real world terms.
Random forests	Although a random forest consists of many decision trees, understanding feature importance is harder compared to a single tree.
Support vector machines	Although a 2D support vector machine can be easier to interpret, higher dimensional applications cannot be visualised and decisions are hard to explain.

** this includes most AI models / LLMs*

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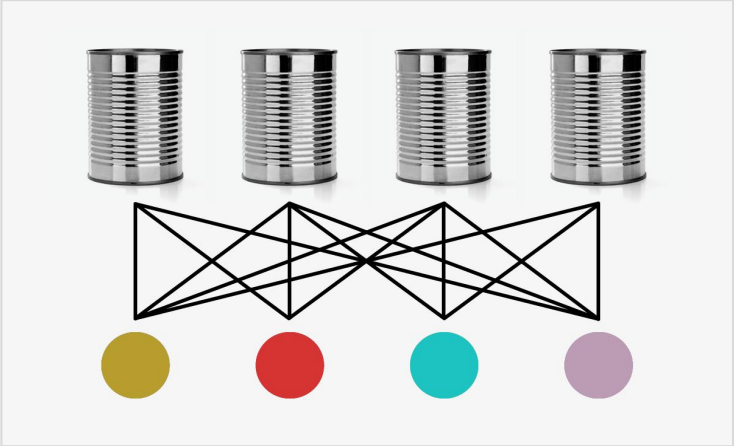
Building Explainable Models

What does explainability look like?

There are different ways of providing explanations, depending on which model is being used.

In this session we'll explore...

- Decision tree visualisations
- Decision tree feature importance
- Partial dependence plots
- Shapely additive explanations (SHAP)



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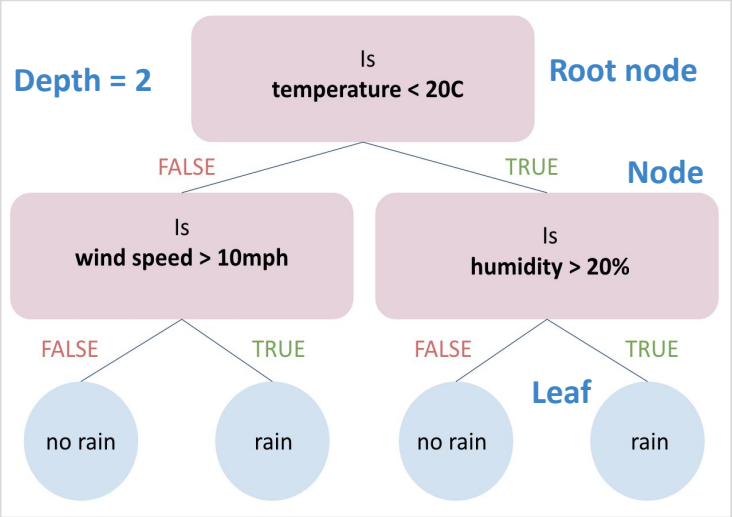
Decision Tree Visualisations

Let's have a quick refresher on decision tree classifiers:

Decision tree classifiers are a supervised, multi-class classification method.

Training a decision tree means optimising two things to give the greatest accuracy on the training dataset:

- The positions of features in the tree
- The thresholds associated with each feature



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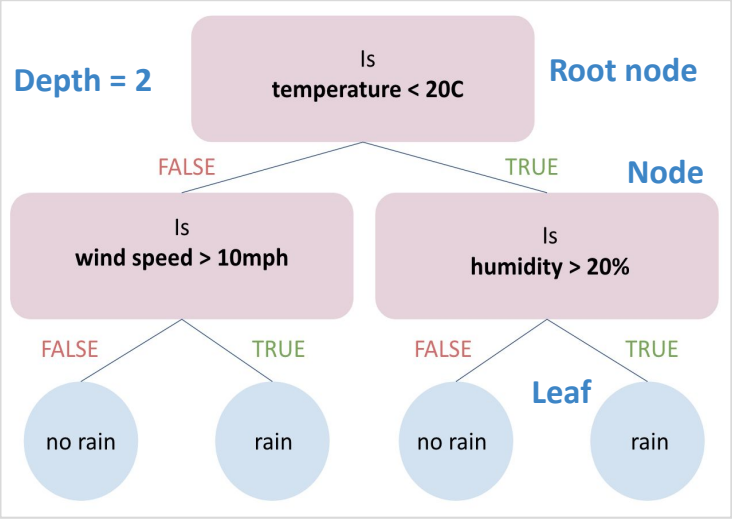
Decision Tree Visualisations

The final, trained decision tree might look something like the one shown. Note:

- The features and their associated thresholds are at the nodes.
- The classifications are at the leaves.

This is a classification method that lends itself to being visualised.



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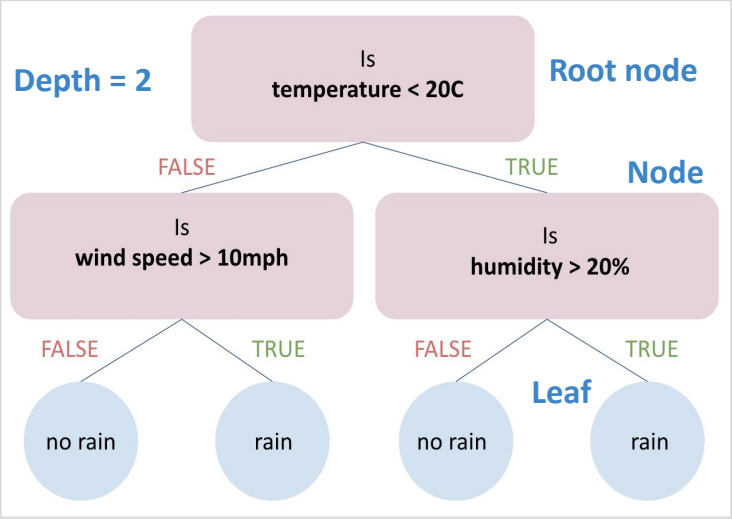
Decision Tree Visualisations

What can we determine from a decision tree visualisation?

- The **more important** features are closer to the top of the tree
- The **thresholds** at each node tell us about the cutoff values for each feature



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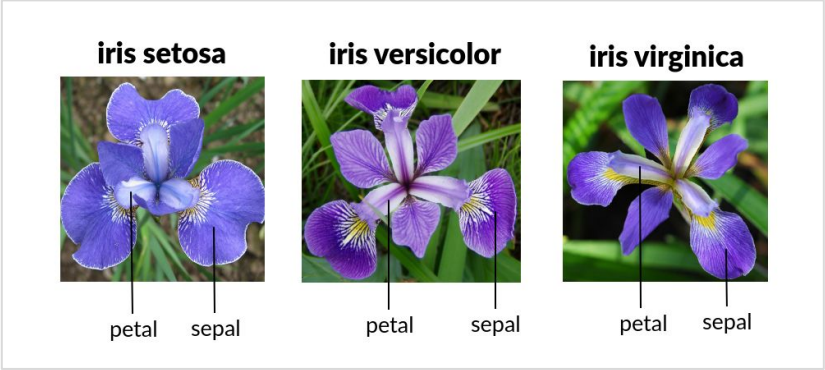
Visualising a decision tree

- ➔ Open **02-model-explainability.ipynb** and go to the section titled '**Visualising a Decision Tree**'
- ➔ We'll build a decision tree to classify iris species using measurements taken from the plant.
- ➔ We will run the code cells and discuss the results together.



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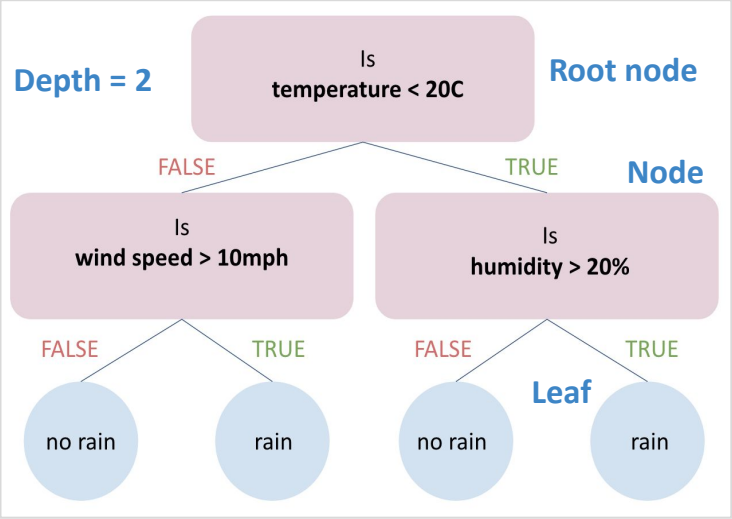
Decision Tree Feature Importance

Decision tree visualisations are a useful qualitative tool.

But would be great to have a quantitative explanation of the relative importance of different features.

That's where **feature importance** comes in.

Let's dip back into our decision tree refresher.



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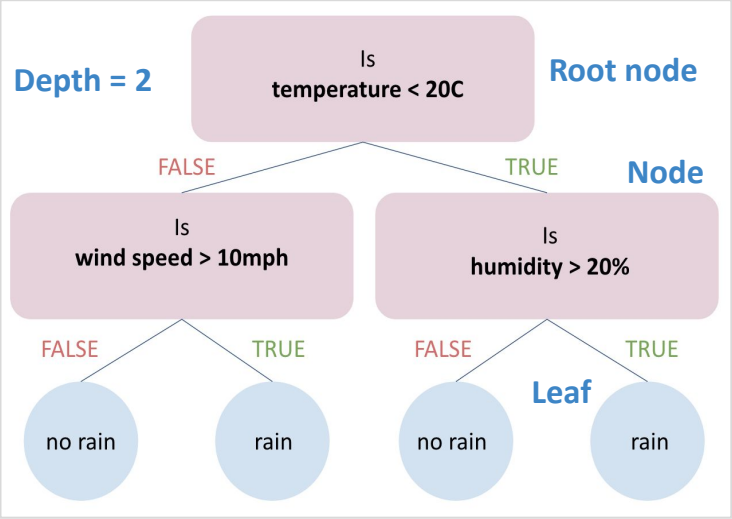
Decision Tree Feature Importance

Remember that training a decision tree involves:

Starting at the root note, and finding the feature + threshold combination that produces the smallest **Gini impurity** in the resulting nodes

Repeating this process at each node until either:

- A pure node (or leaf) is reached that contains only one class of points
- The tree reaches a maximum, user-specified depth



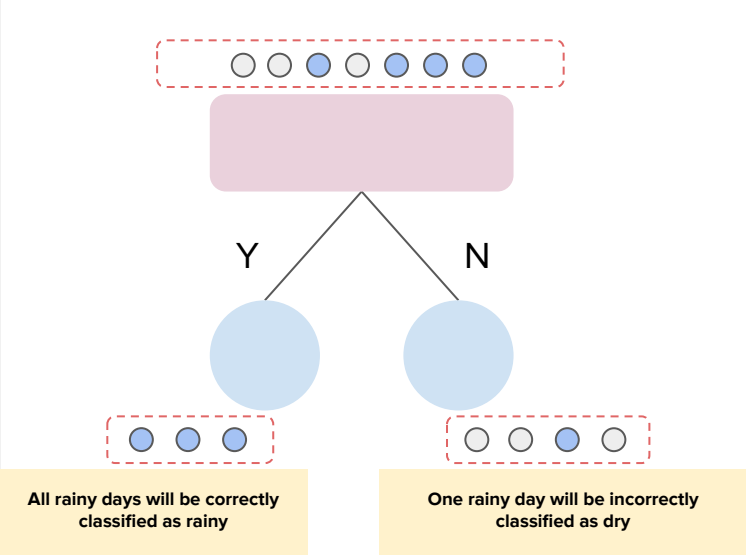
Decision Tree Feature Importance

The **Gini impurity** is a measure of how 'impure' the mix of data points at a node are.

The Gini impurity is high when a node contains a mix of the two classes, and it's low when a node contains mostly points of one class.

If a node contains an important/powerful feature, the impurity **decreases** after the split.

We can see this in the tree on the right.



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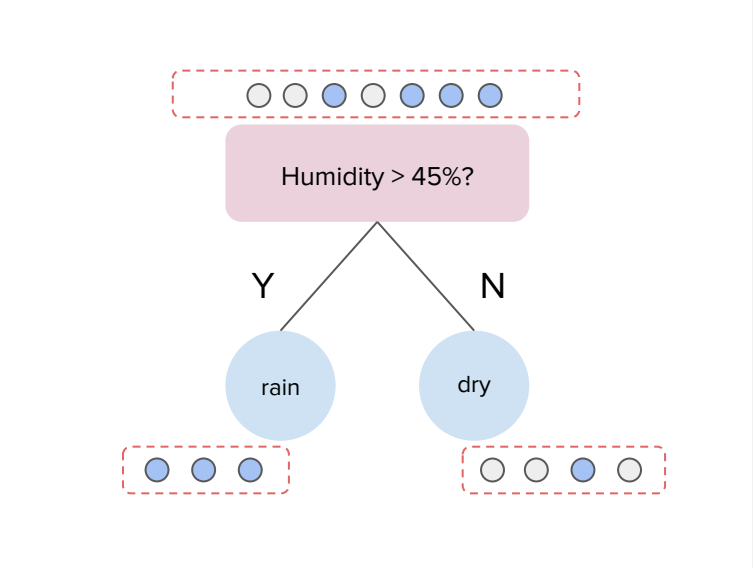
Decision Tree Feature Importance

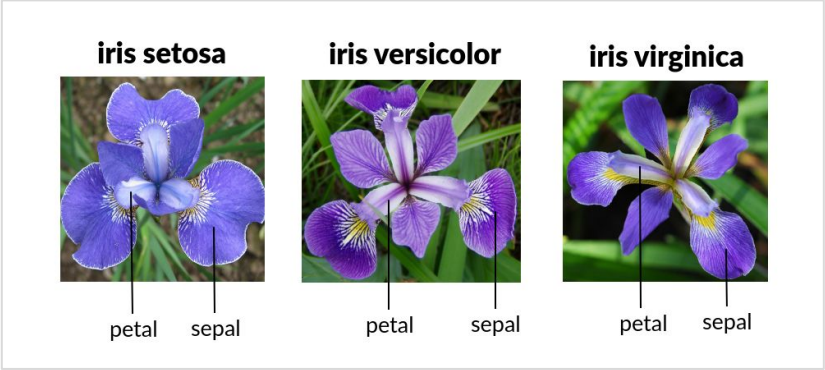
The ‘importance’ of a feature in a tree is the amount it reduces the Gini impurity.

If a feature is more important, it will decrease the Gini impurity by more, because it will produce a cleaner split between the classes.



Feature Importance

- ➔ Open **02-model-explainability.ipynb** and go to the section titled '**Feature Importance**'
- ➔ Let's build on our previous decision tree exercise, to get feature importances.
- ➔ We will run the code cells and discuss the results together.



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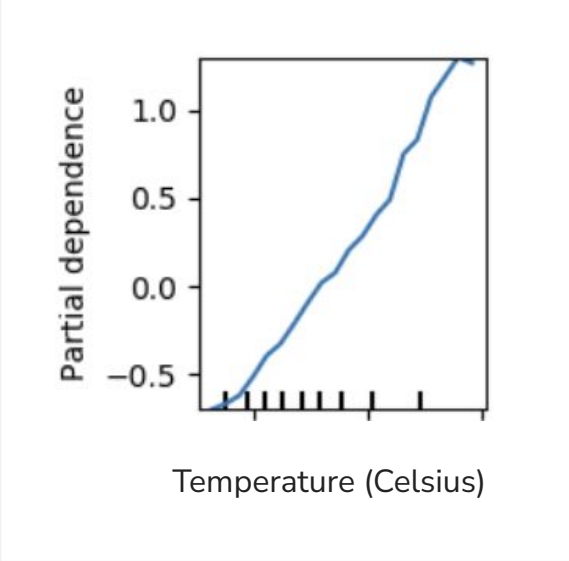
Partial Dependence Plots (PDPs)

Feature importance shows us which features are the strongest in a model.

But what if we wanted to know:

- How does a specific feature influence the model's predictions?
- Is the relationship between a feature and the target linear or non-linear?
- Are there critical points where the model's decision changes abruptly?

Partial dependence plots can help!



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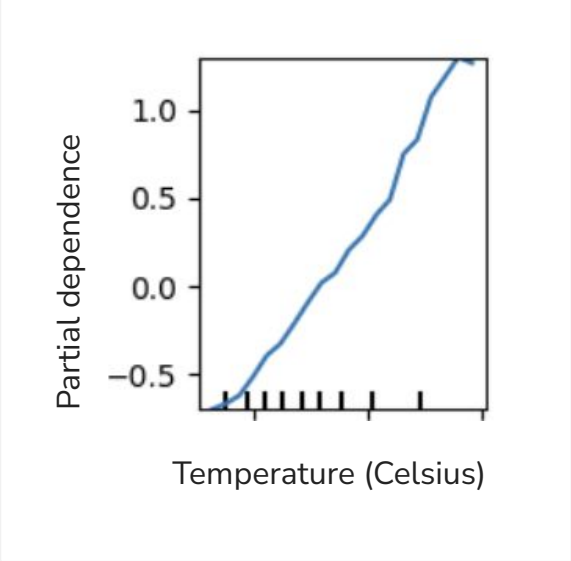
Partial Dependence Plots

A partial dependence plot typically shows how changes in one feature affect the model output.

To produce a plot:

- Choose the feature of interest (e.g. temperature).
- Take all the training examples and replace the chosen feature's value with a fixed value (e.g. 25C).
- Make predictions using the trained model for each point in the modified dataset.
- Compute the average prediction at each feature value.
- Plot these average predictions against the feature values.



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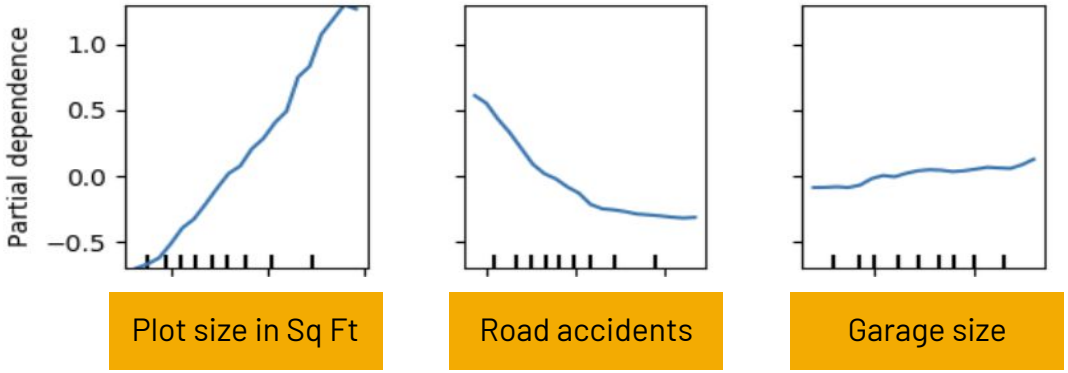
Partial dependence plots

- Imagine you've built a model to predict house prices based on three features: the size of the plot of land, neighbourhood road accidents, and size of garage.
- What do these PDPs tell us about how each of these features affects the target variable?



Model Explainability

Partial dependence plots are straightforward to interpret, based on the shape of the curve.

LINEAR	NON-LINEAR	STEEP-LIKE
A linear PDP suggests that model responds proportionally to changes in the features.	A curve in the PDP indicates diminishing or accelerating effects. A plateau suggests that beyond a certain point, increasing the feature no longer affects predictions.	If the PDP shows sudden jumps, the model makes predictions based on thresholds.

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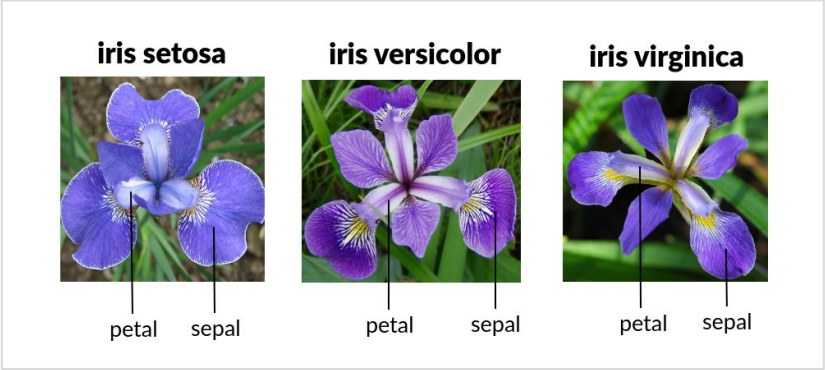
Partial dependence plots

- ➔ Open **02-model-explainability.ipynb** and go to the section titled '**Partial dependence plots**'
- ➔ Let's build on our previous decision tree exercise, to produce PDPs for each feature.
- ➔ We will run the code cells and discuss the results together.



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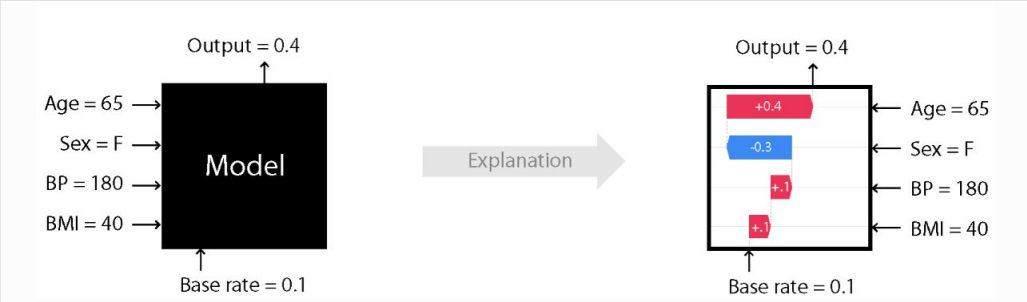
SHapely Additive exPlanations (SHAP)

SHAP is an alternative method for explaining how a model makes predictions.

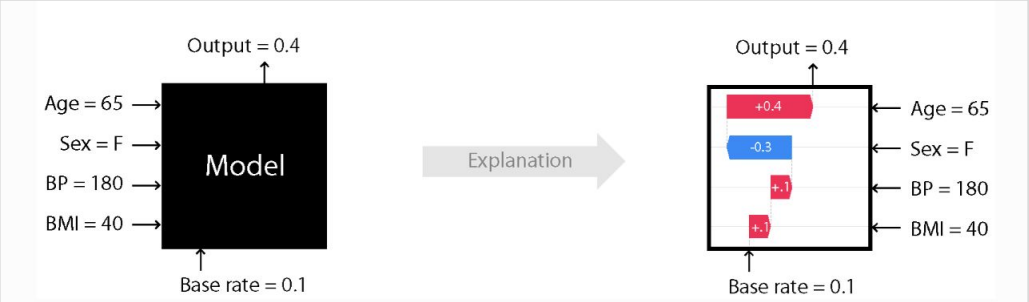
SHAP shows how much each feature contributes to the model's final decision.



SHapely Additive exPlanations (SHAP)

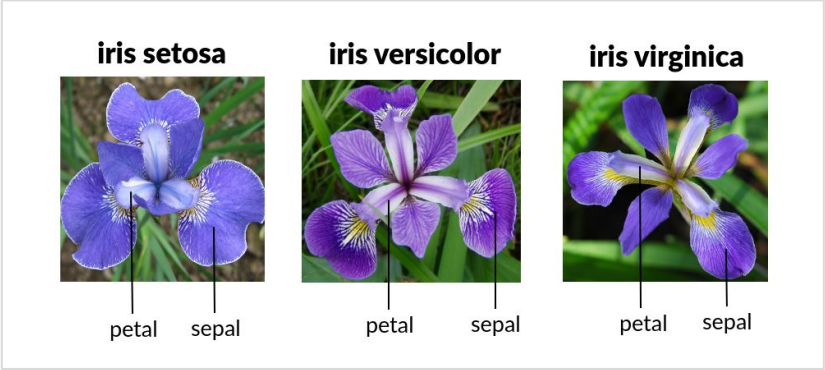
It is based on ‘Shapley values’, a concept from game theory that fairly distributes contributions among players in a game.

In this case, the ‘players’ are the features of your model.



SHAP values

- ➔ Open **02-model-explainability.ipynb** and go to the section titled '**SHAP Values**'
- ➔ Let's build on our previous decision tree exercise, to produce SHAP values.
- ➔ We will run the code cells and discuss the results together.



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Bring It Home

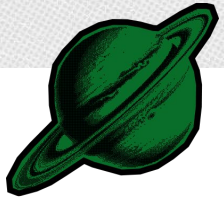
Key takeaways

There are 3 important components of explainability

Transparency, Interpretability and Feature Importance

Explainability is important

It helps with trust, fairness, adhering to regulation, improving models, and reducing ethical risks



There are different ways of providing explanations

For example decision tree visualisations, feature importance, partial dependence plots, and SHAP

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